SIMPLIFIED FORMS AND NORMAL FORMS

A STRAIGHT FORWARD WAY OF IMROVING A GRAMMAR WITHOUT CHANGING LANGUAGE

STEPS

- Eliminating certain types of productions that may be aukward to work with, then
- Standardizing the productions so that they all have a certain normal form

AUKWARD PRODUCTIONS

1. Λ-production (like $T \to \Lambda$)
2. Unit production (Nonterminal $\to$ Single Nonterminal like $T \to A$)

Having Standardized CFG,
It is possible to answer (using algorithm), whether a given string x is in the language generated by grammar. With $|x| = k$, trying all possible sequences of $2k-1$ productions to see if any of them produces x.

EXAMPLE

Consider the Language

$$L = \{ a^i b^j c^k \mid i = j \text{ or } i = k \}$$

CFG:

$$S \to TU \mid V$$
$$T \to aTb \mid \Lambda$$
$$U \to cU \mid \Lambda$$
$$V \to aVc \mid W$$
$$W \to bW \mid \Lambda$$

① $a^i b^j c^k$ $(i = j)$

$$\{ \underbrace{a^n b^n}_{T} \underbrace{c^m}_{U} \mid m \geq 0, n \geq 0 \}$$

$$S \to TU$$
$$T \to aTb \mid \Lambda$$
$$U \to cU \mid \Lambda$$

② $a^i b^j c^k$ $(i = k)$

$$\{ a^n \underbrace{b^m}_{W} c^n \mid m \geq 0, n \geq 0 \}$$

$$V \to aVc \mid W$$
$$W \to bW \mid \Lambda$$

Convert this CFG to Chomsky Normal Form (CNF)

SOLUTION

STEP I : Killing Λ-production (Eliminating Λ-production)

$\boxed{\text{IDENTIFYING NULLABLE VARIABLES :}}$

W, U, T : As $W \to \Lambda$, $U \to \Lambda$, $T \to \Lambda$

V : $V \Rightarrow W \Rightarrow \Lambda$

S : $S \Rightarrow V \Rightarrow \Lambda$ or $S \Rightarrow TU \Rightarrow \Lambda U \Rightarrow \Lambda\Lambda \Rightarrow \Lambda$

S, T, U, V, W are Nullables.

N is nullable if

$N \to \Lambda$

or

$N \stackrel{*}{\Rightarrow} \Lambda$

Identifying productions having nullable on Right side of the productions

Productions with Nullable on R.H.S of grammar:

$S \to TU$: $S \to T$, $S \to U$
$S \to V$: ~~$S \Rightarrow$~~ Nothing to add
$T \to aTb$: $T \to ab$
$U \to cU$: $U \to c$

$V \to aVc$: $V \to ac$
$V \to W$: nothing to add
$W \to bW$: $W \to b$

NOTE
We cannot throw away $\wedge$-production without adding anything

~~after~~
As we have :

$S \to TU|V$
$T \to aTb|\wedge$
$U \to cU|\wedge$
$V \to aVc|W$
$W \to bW|\wedge$

After eliminating $\wedge$-productions

$S \to TU|V|T|U$
$T \to aTb|ab$
$U \to cU|c$
$V \to aVc|W|ac$
$W \to bW|b$

STEP II : Eliminating Unit productions

Unit productions are :

$S \to T$ $\quad V \to W$
$S \to V$
$S \to U$

> **Unit production**
> productions of the form
> Variable $\to$ one variable
> $N \to A$

As we have :

$S \to TU|\underline{V}|\underline{T}|\underline{U}$
$T \to aTb|ab$
$U \to cU|c$
$V \to aVc|\underline{W}|ac$
$W \to bW|b$

After Eliminating Unit production

$S \to TU|\underbrace{aVc|ac|bW|b}_{V}|\underbrace{aTb|ab}_{T}|\underbrace{cU|c}_{U}$
$T \to aTb|ab$
$U \to cU|c$
$V \to aVc|\underbrace{bW|b}_{W}|ac$
$W \to bW|b$

STEP III Converting to Chomsky Normal Form (CNF)

As we have

$S \rightarrow TU \mid aVc \mid ac \mid bW \mid b \mid aTb \mid ab \mid cU \mid c$

$T \rightarrow aTb \mid ab$

$U \rightarrow cU \mid c$

$V \rightarrow aVc \mid bW \mid b \mid ac$

$W \rightarrow bW \mid b$

Chomsky Normal Form

Every production is one of the two types

$N \rightarrow AB$

$N \rightarrow a$

(i) first Convert the above into

$N \rightarrow$ String of only Nonterminals

or $N \rightarrow$ one terminal

then,

(ii) Convert the resultant CFG into

$N \rightarrow$ String of Exactly two Non-terminals

or $N \rightarrow$ one terminal

} required form of CNF

(i) $S \rightarrow TU \mid X_a V X_c \mid X_a X_c \mid X_b W \mid b \mid X_a T X_b \mid X_a X_b \mid X_c U \mid c$

$T \rightarrow X_a T X_b \mid X_a X_b$

$U \rightarrow X_c U \mid c$

$V \rightarrow X_a V X_c \mid X_b W \mid b \mid X_a X_c$

$W \rightarrow X_b W \mid b$

$X_a \rightarrow a$

$X_b \rightarrow b$

$X_c \rightarrow c$

(ii) Now we are ready to Convert it into CNF

$S \rightarrow TU \mid X_a Y_1 \mid X_a X_c \mid X_b W \mid b \mid X_a Y_2 \mid X_a X_b \mid X_c U \mid c$

$Y_2 \rightarrow T X_b$

$Y_1 \rightarrow V X_c$

$T \rightarrow X_a Y_2 \mid X_a X_b$

$U \rightarrow X_c U \mid c$

$V \rightarrow X_a Y_1 \mid X_b W \mid b \mid X_a X_c$

$W \rightarrow X_b W \mid b$

(CNF)

which is required form of Chomsky Normal Form (CNF).

Example #2

Convert the following into CNF.

$S \to ABCBCDA$
$A \to CD$
$B \to Cb$
$C \to a \mid \Lambda$
$D \to bD \mid \Lambda$

Solution

Step I: Eliminating Λ-production

Nullables: C, D (AS $C \to \Lambda$ & $D \to \Lambda$)

A (AS $A \Rightarrow CD \Rightarrow \Lambda D \Rightarrow \Lambda\Lambda \Rightarrow \Lambda$)

Nullables $\boxed{A, C, D}$

Productions with Nullables		Production to add
$D \to bD$		$D \to b$
$B \to Cb$		$B \to b$
$A \to CD$		$A \to C$, $A \to D$, nothing
$S \to ABCBCDA$	(Single)	$S \to BCBCDA$, $S \to ABBCDA$, $S \to ABCBDA$, $S \to ABCBCA$, $S \to ABCBCD$
	(two)	$S \to BBCDA$, $S \to BCBDA$, $S \to BCBCA$, $S \to BCBCD$, $S \to ABBDA$, $S \to ABBCA$, $S \to ABBCD$, $S \to ABCBA$, $S \to ABCBD$, $S \to ABCBCD$
	(three)	$S \to BBDA$, $S \to BCBA$, $S \to BCBD$, $S \to BCBD$ etc.
	(four)	=

So our final CFG has 40 productions including 32-S productions and the ones:

$A \to CD$
$B \to Cb \mid b$
$C \to a$
$D \to bD \mid b$

Example #3

Convert the following algebraic-expression grammar into CNF.

$S \to S+T \mid T$

$T \to T*F \mid F$

$F \to (S) \mid a$

Solution:

Step I . Eliminating Λ-production

No Null production here

Step II : Eliminating Unit production

$S \to S+T \mid T*F \mid (S) \mid a$

$T \to T*F \mid (S) \mid a$

$F \to (S) \mid a$

Step III :

$S \to SP \mid TQ \mid (R \mid a$

$P \to +T$

$Q \to *F$

$R \to S)$ required CNF

$T \to TQ \mid (R \mid a$

$F \to (R \mid a$